

AYUSH PANDEY

ML Engineer | Data Scientist

New Delhi, India | +91 9205426712 | pandeyayush590@gmail.com

PROFESSIONAL SUMMARY

ML Engineer and Data Scientist with 3.5+ years of industry experience at Tata Consultancy Services and a research background from IIT Madras HTIC. Specializes in end-to-end deep learning pipelines (CNNs, Transformers, ViT, YOLO), medical imaging (TB detection, ECG classification, EEG analysis), and competitive ML (Kaggle, DrivenData).

PROFESSIONAL EXPERIENCE

Tata Consultancy Services

New Delhi, India

Data Analyst & ML Engineer

December 2021 – May 2025

- Designed SQL/HiveQL/ImpalaQL queries for client data extraction across Oracle and distributed Hive/Impala systems.
- Built ensemble and boosting classification models (delinquency prediction) achieving 90% test accuracy.
- Developed customer analytics dashboards in Power BI; identified optimal offer strategies increasing customer acquisition by 20%.
- Designed a Hive-based fraud detection database system aggregating failed transaction data from multiple business sources.
- Engineered Python-based ETL pipelines consuming REST APIs with business-rule filtering, reducing data errors by 15%.

Healthcare Technology Innovation Centre — IIT Madras

Chennai, India

Algorithm Developer & Data Science Intern

July 2021 – October 2021

- Developed mathematical models identifying biometrics influenced by exercise; findings drove core feature redesign of the Repose health app.
- Built an ECG-based sleep stage classification model using RNNs and signal processing achieving 87% test accuracy.
- Contributed to ML feature development, supporting a 10% increase in app-purchase revenue.

RESEARCH PROJECTS & COMPETITIVE ML

TB Detection Pipeline | YOLOv8m-seg · Attention U-Net · EfficientNet-B4 · ViT-Small · PyTorch

- Multi-phase medical imaging pipeline: Phase 1 — YOLOv8m-seg for lung segmentation; Phase 2 — Attention U-Net with EfficientNet-B4 + SCSE attention for lesion segmentation (Dice 0.830); Phase 3 — ViT-Small classifier (AUC 0.988).
- Unified ~11K image dataset from Roboflow, SFLTb, and TBX11K sources; addressed severe class imbalance using weighted sampling and focal loss.
- Self-directed research project demonstrating end-to-end proficiency from data engineering to model evaluation.

Dog Muzzle Biometric Re-ID System | ResNet-18 · ArcFace · CBAM · FPN · Triplet Loss · SAM · PyTorch

- Built a dog identity re-identification system using metric learning (triplet loss, ArcFace) with CBAM attention and FPN multi-scale features; full-face crops achieved 96.8% Top-5 accuracy vs. 34.67% for SAM-cropped muzzle inputs.
- Developed Laplacian-based frame sharpness selection for video input preprocessing.
- Invention Disclosure Form in preparation for IP filing on the biometric pipeline.

AMIA Public Challenge 2026 — VinDr-CXR | YOLOv8x · A100 GPU · 14-class pathology detection

- Trained YOLOv8x on VinDr-CXR chest X-ray dataset for 14-class pathology detection; best mAP50 ~0.379 at epoch 26 on NVIDIA A100.
- Active Kaggle competition participation with iterative training and evaluation workflow.

Harmful Brain Activity Classification | *VGG16 · ResNet-3D · EEG · Kaggle / Harvard Medical School*

- Built image-processing model to classify brain seizure patterns in EEG graphs sourced from Harvard Medical School via Kaggle; VGG16 + ResNet-3D achieving KL Divergence loss of 0.45.

Sleep Stage Classification System | *RNN · ECG · Signal Processing*

- Engineered preprocessing pipeline (noise removal, baseline correction) on ECG data; RNN-based classifier achieved 87% accuracy across sleep stages.

VSSUT Satellite Launch Vehicle (VSLV) | *Telemetry · IMU · Python · Avionics*

- Served as Telemetry Lead and Head of AI/ML Department on India's first student-built reusable rocket.
- Built thrust prediction system from engine parameters; improved fuel and motor quality by 45%.
- Developed real-time rocket trajectory tracking system with altitude and IMU monitoring, improving operational efficiency by 30%.

TECHNICAL SKILLS

Languages	Python, SQL, HiveQL, ImpalaQL
ML / DL Frameworks	PyTorch, TensorFlow, Keras, Scikit-Learn
Computer Vision	YOLOv8, U-Net, Attention U-Net, ViT, ResNet, EfficientNet, VGG, CBAM, FPN, ArcFace
LLMs & GenAI	RAG, Ollama, LLaMA, VLMs, Hugging Face, LangChain, LangGraph, LlamaIndex
MLOps & Deployment	MLflow, Docker, Streamlit, Weights & Biases
Vector Databases	Pinecone, FAISS
Data & Visualization	Power BI, Pandas, NumPy, Matplotlib, Seaborn
Platforms	Kaggle, DrivenData, HackerEarth, Google Colab, AWS (basic)

EDUCATION

Veer Surendra Sai University of Technology (VSSUT)

B.Tech — Chemical Engineering

CGPA: 7.29 | First Class Distinction

Burla, Odisha

July 2017 – July 2021

CERTIFICATIONS

- Computational Thinking using Python — MITx on edX
- Machine Learning — Stanford Online (Andrew Ng)
- Combinatorics and Probability — UC San Diego on Coursera

ACHIEVEMENTS & AWARDS

- 1st Runner-up — Tata Crucible Hackathon 2020
- 1st Runner-up — Smart India Hackathon 2020
- Best Research Project — Sristi Summer School 2019
- Active Kaggle competitor — TB detection, BirdCLEF 2026, VinDr-CXR (AMIA 2026), Harmful Brain Activity Classification
- Invention Disclosure Form in progress for dog biometric re-identification system